MODELING ROAD TRAFFIC CRASHES IN KWARA STATE USING POISSON REGRESSION MODEL, NEGATIVE BINOMIAL REGRESSION AND GENERALIZED NEGATIVE BINOMIAL REGRESSION MODEL (2015-2021).

BY

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17/56EG002

BEING A PROJECT SUBMITTED TO THE DEPARTMENT OF STATISTICS IN THE PARTIAL FUFILMENT OF THE OF THE REQUIREMENTS FOR THE AWARD OF BACHELOR OF SCIENCE (HONOURS) DEGREE IN STATISTICS OF THE UNIVERSITY OF ILORIN, ILORIN, NIGERIA.

JANUARY, 2022.

ATTESTATION

This is to certify that the project work entitled **"MODELING ROAD TRAFFIC CRASHES IN KWARA STATE USING POISSON REGRESSION MODEL, NEGATIVE BINOMIAL REGRESSION AND GENERALIZED NEGATIVE BINOMIAL REGRESSION MODEL (2015-2021)"** is an original work carried out by **Abdhamid, Mubarak Alao** with the matriculation number **17/56EG002** under my supervision.

…….	**……**
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CERTIFICATION

This project work has been read and approved as meeting the partial fulfilment for the award of Bachelor of Science (B. Sc.) in Statistics in the Department of Statistics of University of Ilorin, Ilorin, Nigeria.

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DEDICATION

This project is dedicated to Almighty Allah for the strength, wisdom, guidance, ability and understanding he has given to me through all my days as an undergraduate, his guidance during the tough times and also my parents Mr and Mrs. Abdulhameed, my junior brothers and sisters for their incomparable support, advise, assistant and encouragement.

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All praises and adorations belong to the Almighty God. A Project is a set of tasks that turn human idea into something, It is the original work of man but more or less a combination of suggestions, guidance, ideas, reviews, impact and contributions involving many people.

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ABSTRACT

Road traffic accidents is trending one of the challenges that have drawn the attention of researchers, statisticians, road users, families, business men, economy and the general public. Some of these challenges arising from Road traffic crashes include; death of loved one, fractures, loss means of livelihood and other various degrees of injury, therefore, This research was targeted at modeling Road traffic crashes data in Kwara state, Nigeria using Poisson Regression model, while the specific objectives of the study include; assessing factors that contribute to high rate of road traffic crashes. Secondly, to fit an appropriate model to data on road traffic crashes in Kwara state, Nigeria and to examine the already estimated model used in determining and modeling road traffic crashes in Nigeria. The data for the study was a secondary data collected from the policy and research department of the Federal Road Safety commission of Kwara state between January (2015) to June (2021). The results obtained from the analysis revealed that the Poisson regression could not capture over-dispersion. So, other forms of Poisson Regression models such as the Negative Binomial Regression and Generalized Negative Binomial Regression were also used in the estimation. However, the Generalized Negative Binomial Regression Model contains the least Akaike Information Criterion (AIC) based on the selections of the overall best-fitted model. From our results, we can conclude that the first and the second quarter of the year in Kwara state has high rate of road traffic crashes cases which means more attention should be paid to this month than December which people think it is the month

which high cases of road traffic crashes are recorded most. The cases of road traffic crashes are vital when a person is injured or killed as it increases the number of cases recorded.

CHAPTER ONE

INTRODUCTION

1.1 Background of study

Transport is an important element in economic development and it affords the social and political interaction that most people take for granted. The provision of transport infrastructure has grown extensively across the globe through a range of networks of modes which have undergone technological improvements cutting across the motive power, the tracks as well as the means which serve as compartment for passengers and goods. It is also a key player in the transfer and distribution of goods from the input point through the manufacturing line to the customers. Perhaps, this led to the assertion by (Afolabi and Gbadamosi,2007), that there is no escape from transport since it is a key stone to civilization (Afolabi and Gbadamosi,2007).

In Nigeria, Road transport is the dominant mode of movement for both freight and passengers. The impact of the railway has been dwindling and it eventually collapses about a decade ago. Air transport is unavailable to the urban poor while the potentials of water for inland transportation has not been fully exploited (Afolabi and Gbadamosi, 2007).

The consequence effect of the negative externalities of transport is accident with its attendant injuries and fatalities capable of naturalizing its social and economic benefits if not well managed. Traffic fatalities from automobile crashes have been known to be high in developing countries in which Nigeria constitute an integral part despite the much lower vehicle ownership in relation to population strength (Afolabi and Gbadamosi, 2007).

Road traffic crashes are of grave concern as many of them end up taking life. The morbidity and mortality burden is rising in developing countries due to a combination of factors, including rapid motorization, poor road and traffic infrastructure, as well as the behavior of road users (Nantulya and Reich, 2002).

Before 2030, deaths from road traffic accidents (RTAs) would surpass cerebrovascular disease, tuberculosis, and HIV/AIDS. Yet, there is little knowledge on the geographic distribution of RTA severity in Nigeria (Nantulya and Reich, 2002).

The recognition of road traffic crashes as a crisis in Nigeria inspired the establishment of the Federal Road Safety Commission (FRSC). The FRSC was established by the government of the Federal Republic of Nigeria vide Decree 45 of 1988 as amended by Decree 35 of 1992, with effect from 18th February, 1998. The commission was charged with responsibilities of maintaining the safety of traffic infrastructure, educating drivers, motorist and other members of the public generally on the proper use of the highway, preventing and minimizing accident on the highway, organization and administration of road safety in Nigeria (https:www.frsc.gov.ng/establishment).

1.2 Statement of Problem

Road traffic injuries are major causes of death and disability globally, with a disproportionate number occurring in developing countries. Road traffic injuries are currently ranked ninth globally among the leading causes of disability adjusted life years lost, and the ranking is projected to rise to third by 2025 (Olagunju, K. Y. 2018). In 2015, developing countries accounted for more than 85% of all deaths due to road traffic crashes globally and for 96% of all children killed. Moreover, about 90% of the disability adjusted life years, lost worldwide due to road traffic injuries occur in developing countries (Olagunju, K. Y. 2018). The problem is increasing at a fast rate in developing countries due to rapid motorization and other factors. However, public policy responses to this epidemic have been muted at national and international levels. Policy makers need to recognize this growing problem as a public health crisis and design appropriate policy responses (Olagunju, K. Y. 2018).

Road traffic injuries in developing countries particularly affect the productive (working) age group (15-44 years) and children. A developing country is defined as a country that has an annual per capital gross national product (GNP) less than US$9361 (£6456), based on 1998 figures from the World Bank. Most low and middle income countries fall into this category (Famoye, F. et al, 2004). Globally, in 2000, 51% of fatalities and 59% of disability

adjusted life years lost due to road traffic injuries occurred in the productive age group (Famoye, F. et al, 2004). Fatality rates among children are especially high in developing countries. In 2000 the fatality rate for children aged 0-4 years was 29.5 per 100 000 population in South East Asia and low income countries of the western Pacific region, compared with 4.5 deaths per 100 000 population in high income countries. For older children, aged 5-14 years, the fatality rate was 28.1 per 100 000 population in Africa compared with 4.8 for North America, western Pacific countries, and high income countries in Europe (Famoye, F. et al 2004).

Road traffic injuries in developing countries mostly affect pedestrians, passengers, and cyclists—as opposed to drivers, in whom most of the deaths and disabilities in the developed world occur. In the United States, for example, more than 60% of road crash fatalities occur in drivers, whereas drivers make up less than 10% of the deaths due to road traffic injuries in the least motorized countries. In developing countries, where most injuries occur in urban areas, pedestrians, passengers, and cyclists combined account for around 90% of deaths due to road traffic injuries. Urban pedestrians account for 55-70% of deaths (Famoye, F. et al 2004).

The choice of mode of transport in developing countries is often influenced by socioeconomic factors, especially income, In Kenya, for example, 27% of commuters who have no formal education were found to travel on foot, 55% usually used buses or minibuses, and 9% used private cars. By contrast, 81% of people with secondary level

education or above usually travelled in private cars; 19% travelled by bus, and none walked. People with little formal education earn low incomes. For them, the affordable means of transport are walking, travelling by bus or truck, or cycling—all of which expose them to high risks for road traffic injuries (Famoye, F. et al 2004).

People in developing countries are frequently aware of these risks. A regular commuter on the buses in Lagos, Nigeria—which are referred to locally as *danfos*, "flying coffins," or *molue*, "moving morgues"—said, "Many of us know most of the buses are death traps but since we can't afford the expensive taxi fares, we have no choice but to use the buses (Famoye, F. et al 2004).

A lot of causes of road traffic crashes are many which include Driving Under Alcohol/Drug Influence (DAD), Use of Phone While Driving (UPWD), Fatigue (FTQ), Poor Weather (PWR), Sign Light Violation (SLV) among others which are discuss below.

Going over the speed limit: most other countries that suffer from high numbers of road accidents, probably the most prominent cause of trouble on the road is exceeding the speed limit. It is said that over 50% of all crashes in the country are due to this particular factor. There is a reason they say 'slow and steady wins the race', as when you try to reach somewhere quickly by going over the limit, you might end up crashing and end up being late, or worst, not even reaching your destination (https://infoguidenigeria.com/causes-road-accidents-nigeria).

Poor condition of roads: Sometimes, even if you abide by all the rules and drive as carefully as you can, you risk getting into an accident because of many holes in the road. A lot of roads in Nigeria are in terrible state, and the further you get from the big cities, the worse they become. The situation is not helped when this cause is combined with the first one as potholes a high speed are a recipe for disaster (https://infoguidenigeria.com/causes-road-accidents-nigeria).

Overloading: While this is more common for commercial vehicles, some of the regular drivers are also guilty for loading their vehicles with more stuff than they can handle. This results in decreased maneuverability, which means that it gets harder to get out of the harm's way in time. Moreover, it is very harmful for the car in general, which might cause it to malfunction out of the blue in the future (https://infoguidenigeria.com/causes-road-accidents-nigeria).

Bad weather: This is probably one of the most understated reasons for road accidents. A lot of people do not really pay attention to the weather outside when they go for a drive, thus not accounting for, for instance, the slippery roads or other factors that might seem unimportant. Therefore, if a cautious situation arises, they might have trouble getting out of it unscathed, as they might not be able to notice something is wrong until it is too late.

Poor vehicle maintenance: It might surprise you just how many people get on the road when their car is in no condition to do so. While some of the car issues might not go beyond aesthetics, some might be driving without realizing that their brakes are not working. This

endangers both the life of the driver and their passengers, as well as everyone else on the road. Poor condition of a vehicle can cause most harm for vehicles that carry many people or heavy/dangerous loads (https://infoguidenigeria.com/causes-road-accidents-nigeria).

Fatigue: One of the most frequent causes of road accidents in Nigeria and the world is tired drivers. This is particularly common among the long haul truck drivers, as they often have to travel long distances without much chance to rest. However, regular drivers are also guilty of getting behind the wheel not in their best condition. This might seriously affect their driving skills and cognitive functions, thus causing harm for everyone involved.

Bad driving habits: Some drivers have no respect for their fellow man when they get on the road. They think that they are always right and the rest are just getting in their way. Moreover, some even get behind the wheel drunk. Such arrogance sometimes leads to dire consequences, as being overconfident in one's skills might result in a nasty car crash. Therefore, even if you consider yourself to be a top-notch driver, it is best to keep your ego in check to assure the safety of others (https://infoguidenigeria.com/causes-road-accidents-nigeria).

1.3 Purpose of the Study

Transportation safety implies the prevention of accidents and the minimization of accident loss. As Nigerians become more mobile, the possibility of accident resulting in the death of people and the destruction of property on highway becomes more of a critical factor.

The consequences of accident on our road are immense. Accidents cause significant losses to present and future and productive manpower of our country. The important of this research aims at choosing a mathematical model that will help understand and manage road traffic crashes in kwara state, Nigeria. It is envisioned that the results of the study will pave way for more realistic road safety prescriptions.

When modeling count data for characteristics and prediction of an event where the dependent variable are non-negative and integer value, it is appropriate to use Poisson regression model, when data exhibit over dispersion, Poisson regression is not appropriate and negative binomial addresses the issue of over-dispersion by including a dispersion parameter to accommodate the unobserved heterogeneity in the data (chin.H and Quddus, 2003). Various factors such as data clustering and misspecifications of model lead to over dispersion in the data. Another importance of negative binomial is that, it is the mixture of family of Poisson distribution with gamma mixing weight (Greewood. M and Yule. G.U, 2002) .Generalized negative binomial reduces to binomial or negative binomial distribution because the mean and the variance are approximately equal, which is an advantage over the Poisson regression.

1.4 AIMS OF THE STUDY:

The main aim of this research is to model the Road traffic crashes in Kwara state using Poisson regression, Negative Binomial Regression and Generalized Negative Binomial Regression.

1.5 OBJECTIVES

1. To assess the appropriateness and fitness of Poisson regression model to road traffic crashes data.

2. To examine the significance of various determinants of road traffic crashes in kwara state.

3. To examine the pattern of occurrence of Road traffic crashes in kwara state.

1.6 METHODOLOGY

This section was discussed under the following sub-headings; Source of data, model specification.

1.6.1 SOURCE OF DATA

A secondary data of road traffic crashes was extracted from Kwara state command, policy and research department of the Federal Road Safety Commission (FRSC), Nigeria. The data was extracted on a monthly basis between the months of January 2015 to June 2021 making it a total of 78 data point. The variables of interest include season (month of the

year), Number of causes, vehicles involved, Number of people injured, Number of people killed. Poisson regression model will be used to fit the rate of road traffic crashes cases on some factors responsible for road accident in Kwara state.

1.6.2 MODEL SPECIFICATION

In line with the specific objective of this study, the models adopted for this study are Poisson regression, Negative Binomial Regression and Generalized Negative Binomial Regression.

CHAPTER TWO

RESEARCH METHODOLOGY

2.1 Introduction

Road traffic accident can be seen as an occurrence in which one or more vehicles collide with a road barricade, another vehicle, person or animal. Road accidents do not only affect people, but also causes damage or loss of property. The major aim of transportation system is to convey people and goods safely from one place to another. The number of cars, motorcycles and tricycles on the road has brought about the problem of loss of lives and property due to accidents that occur daily (Agbonkhese, O.G. et al). The agency responsible for the enforcement of safety rules on roads in Nigeria is the Federal Road Safety Corps. It was established by the Federal Government of Nigeria via Decree 45 of 1988 as amended by the Decree 35 of 1992, with effect from 18th February, 1988. The Commission was given the responsibility of administration of road safety in Nigeria. The FRSC in 2011 reported that between January 2007 and June 2010 a total of 4,017 truck accidents happened on Nigerian roads with an annual average of 1,148 cases and monthly average of 96 cases (https:www.frsc.com/establishment).

2.2.1 Descriptive Statistics

Descriptive statistics are used to summarize data in an organized manner by describing the relationship between variables in a sample or population. Calculating descriptive statistics represent a vital first step when conducting research and should always occur before making inferential statistical comparisons. Descriptive statistics include types of variables (nominal, ordinal, interval, and ratio) as well as measures of frequency, central tendency, dispersion/variation, and position. Since descriptive statistics condense data into a simpler summary, they enable health-care to assess specific populations in a more manageable form [https://www.formpl.us/blog/descriptive-statistics].

2.2 Over dispersed Data

Over dispersion is the presence of greater variability (statistical dispersion) in a data set than would be expected based on a given statistical model. When the observed variance is higher than the mean of a theoretical model, over dispersion has occurred [https://www.formpl.us/blog/over-dispersed-data].

2.3 The Link Function

A Link function in a generalized linear model maps a non-linear relationship to a linear one, which means you can fit a linear model to the data. It connects the predictors in a model with the expected value of the response (dependent) variable in a linear way. [https://www.formpl.us/blog/link-function].

2.4 Goodness of Fit Tests

The goodness-of-fit test is a statistical hypothesis test to see how well sample data fit a distribution from a population with a normal distribution. Put differently, this test shows if your sample data represents the data you would expect to find in the actual population or if it is somehow skewed. Goodness-of-fit establishes the discrepancy between the observed values and those that would be expected of the model in a normal distribution case. There are multiple methods for determining goodness-of-fit. Some of the most popular methods used in statistics include the chi-square, the Kolmogorov-Smirnov test, the chi-square test, and the log likelihood ratio test.

2.5.1 Categorical Data

Categorical data is a collection of information that is divided into groups. I.e., if an organization or agency is trying to get a biodata of its employees, the resulting data is referred to as categorical. This data is called categorical because it may be grouped according to the variables present in the biodata such as sex, state of residence, etc.

Categorical data can take on numerical values (such as “1” indicating Yes and “2” indicating No), but those numbers don’t have mathematical meaning. One can neither add them together nor subtract them from each other [https://www.formpl.us/blog/categorical-data].

Types of Categorical Data

There are two types of categorical data, namely; the nominal and ordinal data.

1. Nominal Data

This is a type of data used to name variables without providing any numerical value. Coined from the Latin nomenclature “Nomen” (meaning name), this data type is a subcategory of categorical data.

Nominal data is sometimes called “labelled” or “named” data. Examples of nominal data include name, hair color, sex etc.

Mostly collected using surveys or questionnaires, this data type is descriptive, as it sometimes allows respondents the freedom to type in responses. Although this characteristic helps in arriving at better conclusions, it sometimes poses problems for researchers as they have to deal with so much irrelevant data [https://www.formpl.us/blog/categorical-data].

2. Ordinal Data

This is a data type with a set order or scale to it. However, this order does not have a standard scale on which the difference in variables in each scale is measured.

Although mostly classified as categorical data, it is said to exhibit both categorical and numerical data characteristics making it in between. Its classification under categorical data has to do with the fact that it exhibits more categorical data character.

Some ordinal data examples include; Likert scale, interval scale, bug severity, customer satisfaction survey data etc. Each of these examples may have different collection and analysis techniques, but they are all ordinal data [https://www.formpl.us/blog/categorical-data].

2.5.2 Numerical Data

Numerical data is a data type expressed in numbers, rather than natural language description. Sometimes called quantitative data, numerical data is always collected in number form. Numerical data differentiates itself from other number form data types with its ability to carry out arithmetic operations with these numbers.

For example, numerical data of the number of male students and female students in a class may be taken, then added together to get the total number of students in the class. This

characteristic is one of the major ways of identifying numerical data (study.com/what-is-numerical-data).

Types of Numerical Data

Numerical data can take two different forms, namely; discrete data, which represents countable items and continuous data, which represents data measurement. The continuous type of numerical data is further sub-divided into interval and ratio data, which is known to be used for measuring items (study.com/what-is-numerical-data).

1. Discrete Data

Discrete Data represents countable items and can take both numerical and categorical forms, depending on usage. It takes on values that can be grouped into a list, where the list may either be finite or infinite. Whether finite or infinite, discrete data take on counting numbers like 1 to 10 or 1 to infinity, with these groups of numbers being countably finite and countably infinite respectively.

A more practical example of discrete data will be counting the cups of water required to empty a bucket and counting the cups of water required to empty an ocean—the former is finite countable while the latter is infinite countable (study.com/what-is-numerical-data).

2. Continuous Data

This is a type of numerical data which represents measurements—their values are described as intervals on a real number line, rather than take counting numbers. For example, the Cumulative Grade Point Average (CGPA) in a 5 point grading system defines first-class students as those whose CGPA falls under 4.50 - 5.00, second class upper as 3.50 - 4.49, second class lower as 2.50 - 3.49, third class as 1.5 - 2.49, pass as 1.00 - 1.49 and fail as 0.00 - 0.99..

A student may score a point 4.495, 2.125, 3.5 or any possible number from 0 to 5. In this case, the continuous data is regarded as being uncountably finite (study.com/what-is-numerical-data).

2.6 Poisson Regression

The Poisson regression models are generalized linear models (GLM) with logarithm as the link function. In statistics the generalized linear model (GLM) is a flexible generalization of ordinary linear regression that allows for response variables that have error distribution models other than a normal distribution. A generalized linear model is made up of a linear predictor.

$$y_i = \beta_0 + \beta_1 x_{1i} + \cdots + \beta_p x_{pi} + \varepsilon_i \qquad (1)$$

And two functions, A Link function that describes the mean,

$$E(y_i) = \mu_i Depends\ on\ the\ linear\ predictor\ g(\mu_i) = \boldsymbol{y_i} \quad (2)$$

A variance function that describes the variance,

$$V(y_i) Depends\ on\ the\ mean\ V(y_i) = \phi V(\mu) \quad (3)$$

Suppose $y_i \sim poisson,\ Then, E(y_i) = \lambda_i and\ V(y_i) = \lambda_i$ (4)

The GLM generalizes linear regression by allowing the linear model to be related to the response variable via a link function. Link function here is the function that links the linear model in a design matrix and the Poisson distribution function.

Consider a linear regression model given as

$$y_i = \beta_0 + \beta_1 x_1 + \varepsilon_i = 1, \dots, k \quad (5)$$

If $y_i = \beta_0 + \beta_1 x_1 + \varepsilon_i = 1, \dots, k$

XE IR^n is a vector of independent variable then,

$$Y = X\beta + \varepsilon \quad (6)$$

Where X is an n × (k+1) vector of independent variables or predictors, and a column of 1's $\beta\ is\ a$ (k +1) × 1 vector of unknown parameters and ε is an n×1 $vectorof$ random error terms $with$ zero mean .Hence

$$E(Y / X) = x\beta \quad (7)$$

Recall that, for Generalized Linear Models, we use the link function to transform Y:

That is,

$$G(Y) = log_e(Y) \tag{8}$$

Therefore, this can be written more densely as

$$log_e E(X/Y) = x\beta \tag{9}$$

Thus, given a Poisson regression model with parameter β and an input vector x, the predicted mean of the associated Poisson distribution is given by

$$E(X/Y) = e^{X\beta} \tag{10}$$

If y_i, are independent observations with corresponding values x_i of the predicted variables, then β can be estimated by maximum likelihood. The maximum likelihood estimates lack a closed-form expression and must be found by numerical methods. The probability surface for maximum-likelihood Poisson regression is always convex, making Newton-Raphson or other gradient-based methods appropriate estimation techniques.

Therefore, let y_i be the random variable, that takes nonnegative values, i=1,2,…,n, Where n is the number of observations. Since y_i follows a Poisson distribution, the probability mass function is (pmf)

$$P(Y_i = y_i) = \frac{\lambda_i^{y_i} e^{-\lambda_i}}{y_i!}, y_i = 0,1,2,3, \ldots \quad (11)$$

$$P(Y_i = y_i) = f(y) = \frac{e^{-\lambda} \lambda^{y}}{y_i}, y_i = 0,1,2,3, \ldots \quad (12)$$

With mean and variance as

$$E\ (y) = V(y) = \boldsymbol{\lambda} \quad (13)$$

Where the conditional mean or predicted mean of the Poisson distribution is given as

$$E(Y_i) = e^{x\beta} = \lambda_i \quad (14)$$

Where X is the value of the explanatory variable and $\beta = (\beta_1, \beta_2, \ldots, \beta_q)$ is the unknown k-dimensional vector of regression parameter.

The mean of the predicted Poisson distribution is given by

$$E(Y/X) \text{ and variance of } y_i \text{ as } V(Y/X).$$

The parameter β can be estimated by maximum likelihood estimation method as follows;

$$\mathfrak{l}(\beta) = \prod_{i=1}^{n} \frac{\lambda_i^{y_i}\ e^{-\lambda_i}}{y_i!} \quad (15)$$

The log-likelihood function is given by

$$In\,\mathfrak{l}(\beta) = \sum_{i=1}^{n}[-\lambda_i + y_i In\lambda_i - Iny_i!]$$

Substituting $\lambda_i = e^{x\beta}$ in the equation above, we have

$$In\mathfrak{l}\,(\beta) = \sum_{i=1}^{n}[y_i(x\beta) - e^{x\beta} - Iny_i!] \quad (16)$$

Differentiating (16) with respect to β and equating to zero, we have

$$\frac{\partial In\mathfrak{l}(\beta)}{\partial\beta} = \sum_{i=1}^{n}\left[y_i - e^{x\beta}\right]x = 0 \quad i = 0,1,2,\dots,k \quad (17)$$

$$\frac{\partial^2 L}{\partial\beta\partial\beta'} = -\sum_{i=1}^{n}(e^{x_i'\beta}x_i'\,x_i) \quad (18)$$

$$\frac{\partial^2 L}{\partial\beta\partial\beta'} = -\sum_{i=1}^{n}(e^{x_i'\beta}\,x_{ij}x_{ii});\ i,j = 1,2,\dots,p \quad (19)$$

$$E\left(\frac{\partial^2 L}{\partial\beta_j\partial\beta_i'}\right) = -\sum_{i=1}^{n}\left(e^{x_i'\beta}x_{ij}x_{ii}\right);\ i,j = 1,2,\dots,p \quad (20)$$

And the information matrix is;

$$k = \sum_{i=1}^{n} e^{x_i'\beta}\,x_i'x_i \quad (21)$$

$$\frac{\partial^2 L}{\partial\beta_j\partial\beta_i'} = -\sum_{i=1}^{n}(\,e^{x_i'\beta}x_{ij}x_{ii}) \quad (22)$$

$$k_{ji} = E\left(\frac{\partial^2 L}{\partial\beta_j \partial\beta_j'}\right) = -\sum_{i=1}^{n}(e^{x_i'\beta} x_{ij} x_{ii}) \quad (23)$$

$$k_{jir} = \left(\frac{\partial^2 L}{\partial\beta_j \partial\beta_i \partial\beta_r}\right) = -\sum_{i=1}^{n}(e^{x_i'\beta}\, x_{ij} x_{ii}) \quad (24)$$

$$k_{ij}^{(r)} = \frac{\partial k_{ji}}{\partial\beta_r} = -\sum_{i=1}^{n}(e^{x_i'\beta} x_{ij} x_{ii} x_{ir}); i, j, r = 1,2, \ldots., p \quad (25)$$

2.7 Negative Binomial Regression (NBR)

The negative binomial distribution can be viewed as a generalization of the Poisson distribution. The negative binomial can be viewed as a Poisson distribution where the Poisson parameter is itself a random variable, distributed according to a Gamma distribution. Thus the negative binomial distribution is known as Poisson-Gamma mixture. As the most common alternative to Poisson regression, the negative binomial regression addresses the issue of over-dispersion by including a dispersion parameter to accommodate the unobserved heterogeneity in the count data. The negative binomial regression (Poisson-Gamma) can also be considered as generalization of Poisson regression. As its name implied, the negative binomial (Poisson-gamma) is a mixture of two distributions.

The probability mass function for negative binomial regression is

$$f(y_i; \alpha, \mu_i) = \frac{\Gamma(\alpha+y_i)}{\Gamma(\alpha)\,\Gamma(y_i+1)}\left[\frac{\alpha}{\mu_i+\alpha}\right]^{\alpha}\left[\frac{\mu_i}{\mu_{i+}+\alpha}\right]^{y_i} \quad (26)$$

The mean and variance are:

$$E(y_i; \alpha, \mu_i) = \mu_i \quad (27)$$

$$Var(y_i; \alpha, \mu_i) = \mu_i + \frac{\mu_i^2}{\alpha} \quad (28)$$

The next steps consist of defining the log-likelihood function and it can be shown that [3],page 81

$$In\left(\frac{\Gamma(y_i+\alpha^{-1})}{\Gamma(\alpha^{-1})}\right) = \sum_{j=0}^{y-1} In(j + \alpha^{-1}) \quad (29)$$

2.8 Generalized Negative Binomial Regression

The generalized negative binomial distribution was developed by compounding the negative binomial distribution with another parameter which takes into account the variations in the mean and the variance. The parameter is such that both the mean and the variance are positively correlated with the value of the parameter, though the variance increases or decreases faster than the mean. The generalized negative binomial distribution reduces to the binomial or the negative binomial distribution as particular cases and converges to a Poisson-type distribution in which the variance may be more than, equal to or less than the mean, depending upon the value of the parameter. The model is defined as follows;

$$\binom{n+x\beta}{x}\frac{\lceil n+x\beta+1}{x!\ \lceil(n+x\beta-x+1)} \qquad (\mathbf{30})$$

$Taking\ z=\frac{\alpha}{(1-\alpha)}, the\ expression\ given\ a\ genaral\ identity\ as, 1=$

$\sum_{x=0}^{\infty} b_{\beta}(x,n,\alpha), then$

$$\beta_{\beta}(x,n,\alpha)=\frac{n\lceil n+x\beta}{x!\lceil(n+x\beta-x+1)}\alpha^{x}(1-\alpha)^{n+\beta x-x}, n>0, x=$$

1,2,3, … (31)

$$b_{\beta}(x,n,\alpha)=0, \forall x\leq n\ such\ that\ n+\beta_{m}<0$$

Where $\sum(n)=\ \lambda\,(n+1)=a\,(b+n+1)$, the function $g_n(x)=f_{NB}(n;b+1,e^{ax}), n=$ 0,1,2, ... can be thought as the solution of chapman Kolmogorov differential equation with transition rate as $\sum(n)=n\geq 0$ that linking increasing from the previous comment results proved by [5] then the solution of the system of the differential equation is given by

$$P_{l,n(x)}=g_{n-1}(x)=f_{NB}(n-1;b+1,e^{-ax})\binom{b+n-1}{n-1}(e^{-ax})^{b+1}(1-e^{-ax})^{n-1}\forall, n\leq 1$$

(32)

$$P\{Y(t)=n\}\int_0^1\binom{b+n-1}{n-1}\left[e^{-a(t-\tau)b+1}\right][1-e^{-a(t-\tau)}]^{n-1}\lambda_0 e^{-\lambda_0}dt \qquad (33)$$

$$=\binom{b+n-1}{n-1}\lambda_0\int_0^t e^{-a(t-\tau)(b-1)}e^{-\lambda_0 t}[1-e^{-a(t-\tau)}]^{n-1}dt \qquad (34)$$

And substituting x = t in the last integral it follows that;

$$P\{Y(t)=n\} = \binom{b+n-1}{n-1}\lambda_0 e^{-\lambda_0 t}\int_0^t e^{(\lambda_0 - a(b+1))x}(1-e^{-ax})^{n-1}dx$$

(35)

$$= \binom{b+n-1}{n-1}\lambda_0 e^{-\lambda_0 t}\int_0^t (e^{-ax})^{b+1-\lambda_0/a}(1-e^{-ax})^{n-1}dx \qquad (36)$$

Finally consider $y = e^{-ax}$ in the integral of $n \geq 1$ that $P\{Y(t)=n\} =$

$$\binom{b+n-1}{n-1}\frac{\lambda_0 e^{-\lambda_0 t}}{a}\int_{e^{-at}}^1 y^{b-\lambda_0/a[1-y]^{n-1}} \qquad (37)$$

Hence the probability distribution function of the generalized negative binomial model is determined by

$$P\{Y(t)=0\} = e^{-t\lambda_0} and\ P\{Y(t)=n\} =$$

$$\binom{b+n-1}{n-1}\frac{\lambda_0 e^{-\lambda_0 t}}{a}\int_{e^{-at}}^1 y^{b-\lambda_0/a[1-y]^{n-1}} \qquad (38)$$

$$\text{Mean} = n\alpha[1-\beta\alpha+(\beta\alpha)^2+\cdots] = \frac{n\alpha}{(1-\beta\alpha)} \qquad (39)$$

$$\mu_2 = \frac{n\alpha(1-\alpha)}{(1-\beta\alpha)^3} \qquad (40)$$

Consider $y_1, y_2, \ldots, y_n$ a random sample whose distribution is given by;

$$L(\lambda_0, a, b) = \prod_{i-1}^n P(Y_i = y_i) \qquad (41)$$

$$\prod_{i:y_i=0} P(Y_i = y_i)\prod_{i:y_i>0} P(Y_i = y_i) \qquad (42)$$

$$\prod_{i:y_i>0} e^{-\lambda_0} \prod_{i-1}^{n} \binom{b+y_i-1}{y_i-1} \frac{\lambda_0 e^{-\lambda_0 t}}{a} \int_{e^{-at}}^{1} x^{b-\lambda_0/a[1-x]^{y_i-1}} dx \quad (43)$$

$$\frac{e^{-n_0\lambda_0}\lambda_0{}^{(n-n_0)}e^{-(n-n_0)\lambda_0}}{a^{n-n_0}} \times \prod_{i:y_i.>0}^{n} \binom{b+y_i-1}{y_i-1} \frac{\lambda_0 e^{-\lambda_0 t}}{a} \int_{e^{-at}}^{1} x^{b-\lambda_0/a[1-x]^{y_i-1}} dx \quad (44)$$

$$\frac{e^{-n_0\lambda_0}\lambda_0{}^{(n-n_0)}}{a^{n-n_0}} \times \prod_{i:y_i>0}^{n} \binom{b+y_i-1}{y_i-1} \frac{\lambda_0 e^{-\lambda_0 t}}{a} \int_{e^{-at}}^{1} x^{b-\lambda_0/a[1-x]^{y_i-1}} dx \quad (45)$$

Where $y_i, y_2, \ldots, y_n$ are observed value of $Y(1), n_0 = \sum_{j=1}^{n} 1(0) y_i$ is the number of zero in the sample.

$\lambda_0 > 0, a > 0 \; and \; b = -1$ Thus the likelihood function is obtained by $l(\lambda_0, a, b) =$

$n\lambda_0 + (n - n_0)(\log(\lambda_0 - \log(a)) +$

$\sum \log \binom{b+y_i-1}{y_i-1} \int_{e^{-at}}^{1} x^{b-\lambda_0/a[1-x]^{y_i-1}} \, dx \quad (46)$

To improve the performance of the maximization and logarithm used to obtain the maximum likelihood estimation the following parameterization will be considered $l_0 = log\lambda_0, l_a = \log(a), l_b = \log(b+1)$ hence the likelihood function for the parameterization is expressed as follows

$l(l_0, l_a, l_b) = -ne^{l_0} + (n - n_0)(l_0 - l_a) \sum_{i:y_i>0} \binom{e^{l_b}+y_i-x}{y_i-1} \int_{\exp(-e^{l}a)}^{1} x^{e^{l_b}-e^{-l_0-l_a-1}} (1 -$

$x)^{y_i-1} \, dx \quad (47)$

By substituting equation (29) into (26), the log-likelihood can be computed using the equation

$$L = \sum_{i=1}^{n}[in[\ulcorner(y_i - \alpha^{-1})] - In[\ulcorner(y_i + 1) - \alpha^{-1}In(1 + \alpha\mu_i) - y_i In(1 + \alpha\mu_i) + y_i In(\alpha) + y_i In(\mu_i)] \quad (48)$$

Therefore, the log-likelihood function become [3], page 81)

$$L = \sum_{j=0}^{y_i - 1} In\,(j + \alpha^{-1}) - In(\ulcorner(y_i + 1) - (y_i + \alpha^{-1})In(1 + \alpha\mu_i) + y_i In(\mu_i) + y_i In(\alpha) \quad (49)$$

2.9 Deviance

The deviance is twice the difference between the maximum log-likelihood and the log-likelihood of the fitted model. In multiple regression under normality, the deviance is the residual sum of square. In the case of negative binomial regression, the deviance is a generalization of the sum of squares. The maximum possible log-likelihood is computed by replacing $\mu_i\ with\ y_i$ in the likelihood formula. Thus, we have

$$D = 2[l(y_i) - L(\mu_i)]$$

$$2\sum_{i=1}^{n}\{y_i In\left(\frac{y_i}{\mu_i}\right) - (y_i + \alpha^{-1})\ In\ (\frac{1+\alpha y_i}{1+\alpha\mu_i})$$

2.10 Multicollinearity Test

In statistics muticollinearity is a phenomenon in which one predictor variable in multiple regression model can be linearly predicted from the other with a substantial degree of accuracy. Multicollinearity occur when there are high correlation between two or more

predictor variable, which skewing the result in regression model. One of the formal ways of detecting muticollinearity is by the use of the variance inflation factor (VIF).VIF is the quotient of the variance in a model with multiple term.it qualifies the severity of muticollinearity in the least square regression analysis. A VIF value of 10 and above indicates a multicollinearity problem.

$$VIF = \frac{1}{1-R_j{}^2}$$

Where $R_j{}^2$ is the coefficient of determination of a regression of explanatory variable j on all the other explanatory variables

Tolerance is a useful tool for diagnosing muticollinearity which happen when variable are too closely related, and tolerance is associated with each independence variable and range from 0 to 1.Allison (1990) noted that there is not a strict cutoff for tolerance, but suggests a tolerance below 0.40 is cause for concern. Any value under 0.20 suggest serious muticollinearity in the model.

High tolerance (e.g. 0.84) =low multicollinearity

Low tolerance (e.g. 0.19) =high muticollinearity

2.11 MODEL SELECTION TEST

2.11.1 Akaike Information criterion (AIC)

The Akaike information criterion (AIC) is a measure of the relative quality of a statistical model for a given set of data. That is given a collection of model for the data, AIC estimates the quality of each model, relative to each of the model. Hence AIC provides a mean for model selection, given a set of candidate models for the data, the preferred model is the one with the minimum AIC value i.e the smaller the AIC value, the better the model).it was also mention that Akaike Information criterion is one of the most commonly used fit statistics. It has a formula

AIC (l) = -2(L-K)

AIC (n) =2/n (L-K)

Note that K is the number of predictors including the intercept, AIC is usually outputted by statistical software application, L is the maximized value of the likelihood function for the model.

2.11.2 Bayesian Information criterion (BIC)

Bayesian information criterion is another common fit statistic. It has three formula

BIC(R) =D-(df) In (n)

BIC (L) =2L+Kin (n)

BIC (Q) =2/n(L-K in(k))

Where; df is the residual degree of freedom and L represents the log likelihood.

CHAPTER THREE

DATA PRESENTATION, ANALYSIS AND INTERPRETATION

INTRODUCTION

The data used for this project was a secondary data extracted from the monthly report of road traffic crashes cases in Kwara state from the policy and research department of the Federal Road Safety Commission. We analyze the extracted data trying to understand the factors that influence the rate of road traffic cases in Kwara state. When dealing with count dependent variable, Poisson Regression is the best method to analyze the data, provided the mean of the dependent variable is equal to the variance of the dependent variable or the deviance is closer to 1, this shows that over dispersion is not present in the model, otherwise model like Negative Binomial and Quassi Poisson are adopted to remedy the presence of over dispersion.

3.1 DATA PRESENTATION

The Causes of accidents as classified by the Federal Road Safety Commission in Nigeria, are listed below;

1. Speed Violation (SPV)

2. Loss of Control (LOC)

3. Dangerous Driving (DGD)

4. Tyre Burst (TBT)

5. Brake Failure (BFL)

6. Wrongful Overtaking (WOT)

7. Route Violation (RTV)

8. Mechanically Deficient Vehicle (MDV)

9. Bad Road (BRD)

10. Road Obstruction Violation (OBS)

11. Dangerous Overtaking (DOT)

12. Overloading (OVL)

13. Sleeping on Steering (SOS)

14. Driving Under Alcohol/Drug Influence (DAD)

15. Use of Phone While Driving (UPWD)

16. Fatigue (FTQ)

17. Poor Weather (PWR)

18. Sign Light Violation (SLV)

19. Others

An accident could be by one or multiple causes as listed above. Where 0 indicate no causes where 1 is indicated in the data presentation, it means the cause of the accident is one and could be either of the causes listed above. Where 2 is indicated in the data presentation, it means that there are 2 causes of accidents and can be any two causes listed above and so on.

The seasons represent the month in which the cases are recorded and they are categorized and decode as;

January=0 base Month, February=1, March=2, April=3, May=4, June =5, July =6, August=7

September=8, October=9, November=10, December=11.

The data is presented as in table 3.1.1 below:

Cases	**Season (month of the year)**	**Number of causes**	**Vehicles involved**	**No Injured**	**No Killed**
10	**1**	**4**	**16**	**42**	**4**
14	**2**	**5**	**20**	**39**	**4**

19	**3**	**5**	**26**	**52**	**7**
25	**4**	**4**	**31**	**66**	**3**
16	**5**	**6**	**23**	**34**	**9**
11	**6**	**4**	**31**	**45**	**4**
15	**7**	**3**	**13**	**30**	**14**
30	**8**	**4**	**23**	**27**	**4**
35	**9**	**6**	**20**	**57**	**15**
6	**10**	**6**	**22**	**38**	**0**
13	**11**	**3**	**15**	**21**	**2**
14	**12**	**5**	**27**	**55**	**14**
8	**1**	**4**	**12**	**45**	**9**
5	**2**	**4**	**11**	**37**	**5**
5	**3**	**3**	**6**	**29**	**3**
6	**4**	**5**	**8**	**6**	**3**
9	**5**	**6**	**12**	**12**	**4**
9	**6**	**6**	**11**	**32**	**3**
5	**7**	**2**	**5**	**20**	**6**
3	**8**	**2**	**5**	**46**	**1**
5	**9**	**5**	**8**	**19**	**2**
6	**10**	**4**	**7**	**23**	**5**

12	**11**	**3**	**19**	**27**	**2**
10	**12**	**4**	**16**	**45**	**25**
11	**1**	**7**	**16**	**77**	**14**
16	**2**	**10**	**23**	**35**	**7**
13	**3**	**9**	**22**	**38**	**7**
16	**4**	**8**	**25**	**80**	**14**
14	**5**	**8**	**25**	**80**	**7**
13	**6**	**9**	**21**	**90**	**13**
17	**7**	**10**	**29**	**92**	**19**
15	**8**	**8**	**24**	**44**	**3**
15	**9**	**8**	**23**	**30**	**10**
17	**10**	**12**	**26**	**42**	**15**
12	**11**	**6**	**12**	**38**	**7**
20	**12**	**7**	**29**	**92**	**16**
7	**1**	**7**	**46**	**91**	**14**
0	**2**	**0**	**0**	**0**	**0**
20	**3**	**9**	**26**	**48**	**2**
21	**4**	**6**	**38**	**53**	**16**
24	**5**	**8**	**29**	**91**	**16**
0	**6**	**0**	**0**	**0**	**0**

16	**7**	**7**	**27**	**33**	**12**
18	**8**	**8**	**26**	**68**	**6**
19	**9**	**7**	**26**	**57**	**17**
18	**10**	**7**	**26**	**52**	**14**
23	**11**	**6**	**36**	**98**	**28**
26	**12**	**10**	**46**	**91**	**15**
19	**1**	**5**	**23**	**73**	**13**
18	**2**	**8**	**28**	**52**	**8**
26	**3**	**9**	**38**	**131**	**16**
27	**4**	**11**	**34**	**153**	**13**
34	**5**	**7**	**53**	**159**	**21**
29	**6**	**7**	**32**	**115**	**20**
23	**7**	**7**	**31**	**98**	**17**
27	**8**	**6**	**36**	**89**	**29**
0	**9**	**5**	**0**	**0**	**0**
0	**10**	**0**	**0**	**0**	**0**
12	**11**	**8**	**20**	**33**	**14**
29	**12**	**11**	**37**	**85**	**29**
36	**1**	**9**	**43**	**129**	**29**
26	**2**	**12**	**32**	**124**	**20**

38	**3**	**4**	**67**	**116**	**34**
19	**4**	**5**	**24**	**145**	**33**
17	**5**	**7**	**41**	**92**	**13**
26	**6**	**10**	**35**	**95**	**31**
28	**7**	**6**	**32**	**55**	**11**
27	**8**	**6**	**38**	**58**	**10**
40	**9**	**8**	**60**	**107**	**37**
21	**10**	**7**	**31**	**80**	**10**
32	**11**	**6**	**47**	**85**	**21**
14	**12**	**10**	**12**	**60**	**16**
32	**1**	**7**	**41**	**77**	**46**
34	**2**	**9**	**49**	**114**	**21**
33	**3**	**7**	**50**	**177**	**8**
44	**4**	**7**	**62**	**158**	**48**
32	**5**	**6**	**41**	**115**	**23**
31	**6**	**5**	**41**	**93**	**9**

SOURCES:

The data was extracted from the monthly report of the Federal Road Safety Commission Kwara state command (policy and research department) between the month of

January2015 to June2021 making a total of 78 data point and the variables needed for the research after multicollinearity test are listed above.

3.2 DATA ANALYSIS

Table 3. 1: Descriptive statistics of the continuous variable

Descriptive Statistics

	N	Minimum	Maximum	Mean	Std. Deviation	Variance
number of monthly cases	78	0	44	18.41	10.404	108.245
Number of causes	78	0	12	6.35	2.593	6.723
vehicles involved	78	0	67	26.49	14.685	215.656
number of injured	78	0	177	65.83	41.033	1683.699
number of people killed	78	0	48	13.08	10.585	112.046

Table 1 shows a clear picture of the descriptive statistics of the continuous variables, with an average of 18 cases per month, while the causes is of average of 6 causes per accident. The number of vehicles involved in accident is an average of 14 vehicles per month, average of 41 people were recorded injured monthly and average of 13 deaths occur every month.

Table 3. 2: Test of Distribution

Hypothesis

Null Hypothesis, H_0: The data does not follow a Poisson Distribution

Alternative Hypothesis, H_0: The data follows a Poisson Distribution

Decision Rule: Reject H_0, if p-value < level of significance (α), otherwise do not reject.

One-Sample Kolmogorov-Smirnov Test

		number of monthly cases
N		78
Poisson Parameter	Mean	18.41
Most Extreme Differences	Absolute	0.240
	Positive	0.217
	Negative	-0.240
Kolmogorov-Smirnov Z		2.118
p-value		0.000

Table 2 shows the one sample Kolmogorov Smirnov test, It indicate the test of distribution, It is to know whether the dependent variable (Number of monthly cases of road traffic crashes) follows a Poisson distribution, the p- value (0.000) which is lesser than 0.05

implies that the null hypothesis is rejected therefore we can conclude that the data follows a Poisson distribution.

Table 3. 3: Multicollinearity Diagonistics

Model		Tolerance	VIF
1	Season(month of the year)	0.911	1.098
	Number of causes	0.723	1.382
	vehicles involved	0.319	3.137
	number of injured	0.303	3.305
	number of people killed	0.471	2.124

Table 3 shows that all the variables have VIF values < 10, using rule of thumb since they are all less than 10, there is no Multicollinearity. Thus all the variable can be Included in the subsequent analysis and model with the Poisson regression, Negative Binomial regression and Generalized Negative Binomial Regression.

Table 3. 4: Omnibus Test

Omnibus Test

Likelihood Ratio Chi-Square	Df	p-value
58454.269	14	0.000

Dependent Variable: number of monthly cases of road traffic crashes

Table 4 elicit the omnibus test and test whether our model incorporate all our predictors and represent a significant improvement in fit over a null model with no prediction, which could also be referred to as an intercept only model. Since the likelihood ratio chi-square test is statistically significant with p-value (0.000) <0.05.it tells us that there is a significant improvement in fitted model over a null model with no predictor, i.e. an intercept only model.

Table 3. 5: Parameter estimate for Poisson regression model

Coefficients	Estimate	Std. Error	z value	P-value	EXP(β)
(Intercept)	-35.413592	0.503481	-54.03	<2e-16	4.169388e-16
Season(Feb)	1.484218	0.202097	39.93	<2e-16	4.4115
Season(march)	1.159931	0.137743	47.96	<2e-16	3.1897
Season(April)	6.158900	0.157385	36.62	<2e-16	472.9076
Season(may)	4.537746	0.282622	24.23	<2e-16	93.4799
Season(June)	4.965548	0.274903	46.25	<2e-16	143.3871
Season(July)	8.287181	0.335155	28.03	<2e-16	3972.6196
Season(Aug)	2.419091	0.205369	43.66	<2e-16	11.2356
Season(Sep)	2.352017	0.262989	40.26	<2e-16	10.5067
Season(Oct)	5.065340	0.226636	24.66	<2e-16	158.4343
Season(Nov)	4.559651	0.140147	42.18	<2e-16	95.5501
Season(Dec)	-4.427757	0.206986	21.39	<2e-16	0.0119
Number of causes.	1.678881	0.051649	32.51	<2e-16	5.3596
No Injured	0.134534	0.002136	62.99	<2e-16	1.1440
No Killed	0.693683	0.008535	81.28	<2e-16	2.0011

Table 3. 6: Deviance, AIC and BIC value for Poisson regression

Parameter	Value
Residual deviance	25516
AIC	25890
BIC	25925.85

Table 7 shows the deviance value, AIC value and BIC value are stated above as 255515, 25890 and 25925.85 respectively.

Table 3. 7: Goodness of Fit Test

	Value	Df	Value/df
Deviance	25516.242	63	405.020
Scaled Deviance	25516.242	63	
Pearson Chi-Square	957497149422067580.000	63	15198367451143930.000
Scaled Pearson Chi-Square	957497149422067580.000	63	
Log Likelihood	-12930.755		
Akaike's Information Criterion (AIC)	25891.510		
Finite Sample Corrected AIC (AICC)	25899.252		
Bayesian Information Criterion (BIC)	25926.860		
Consistent AIC (CAIC)	25941.860		

Table 5 shows the deviance value and discovered that there was dispersion in the data when using the Poisson Regression with the link function identity, because the value of the dispersion is greater than 1.and according to the rule, when the value of the dispersion is close or equal to 1, then there is no trace of over dispersion but when the value is greater than 1 then there is present of over dispersion. Furthermore, the value of deviance (405) is greater than 1, so we can conclude that in estimating road traffic crashes using Poisson regression models, the degree of covariates for the various cases of road traffic crashes in Kwara state suffered from over-dispersion. According to Ayunanda, et al (2013), Over-dispersion usually leads models into producing biased parameter estimates. The consequences of over-dispersion is that it makes the value of the estimated standard error to be wrong, the mean value of the model is not equal to its variance as one of the precondition for Poisson regression models. Subsequently, this may lead to errors in drawing inference about the parameters of the model. To overcome these challenges of modeling counts data, we then resort to the use of negative binomial Poisson regression and the generalized negative binomial regression, since both models can accommodate and capture dispersion parameter in modeling counts data. Since there is presence of over dispersion in the model fitted above which was calculated by dividing the deviance value by the df (degree of freedom) as $\frac{deviance}{n-p} = \frac{25515}{63} = 405$, the rule says, if the value is closer to 1 then there is no dispersion otherwise there is presence of dispersion in the model. As a result of dispersion in the model, a model to remedy the dispersion cases is needed so

therefore a negative binomial model and generalized negative binomial regression were fitted.

Table 3. 8: Parameter Estimate for Negative Binomial Regression Model

Coefficients:	Estimate	Std. Error	z value	P-value	EXP(β)
(Intercept)	1.576126	0.221868	7.104	1.21e-12	4.8361823
Season(Feb)	0.317029	0.228890	1.385	0.16603	1.3730423
Season(March)	-0.066291	0.242273	-0.274	0.78437	0.9358581
Season(April)	0.232825	0.226310	1.029	0.30358	1.2621605
Season(May)	0.511509	0.226814	2.255	0.02412	1.6678053
Season(June)	0.266034	0.226331	1.175	0.23983	1.3047788
Season(July)	0.073267	0.223289	0.328	0.74282	1.0760173
Season(Aug)	0.155148	0.221127	0.702	0.48291	1.1678308
Season(Sep)	0.076092	0.221412	0.344	0.73110	1.0790618
Season(Oct)	0.233562	0.225753	1.035	0.30086	1.2630911
Season(Nov)	0.026365	0.223864	0.118	0.90625	1.0267157
Season(Dec)	-0.058153	0.220009	-0.264	0.79153	0.9435055
Number. of causes	0.062772	0.021428	2.929	0.00340	1.0647842
No Injured	0.006711	0.001699	3.950	7.80e-05	1.0067333
No Killed	0.017736	0.005764	3.077	0.00209	1.0178942

Table 3. 9: Deviance, AIC and BIC value for Negative Binomial Regression

Parameter	Value
Residual deviance	102.4979
AIC	551.487
BIC	589.1943

Table 9 show a clear picture of the AIC value (102.4979), BIC value (551.487), Residual deviance value (102.4979) for Negative binomial Regression.

Table 3. 10: Parameter Estimate for Generalized Negative Binomial Model

Coefficients:	Estimate	Std. Error	z value	P-value	EXP(β)
(Intercept)	2.05347061	0.06124526	33.5286	< 2.2e-16	7.79490731
Season(Feb)	0.21403525	0.10304417	2.0771	0.018895	1.23866631
Season(March)	0.09111571	0.11465035	0.7947	0.213386	1.09539575
Season(April)	0.11978182	0.10212882	1.1729	0.120428	1.12725088
Season(May)	0.17043337	0.11774179	1.4475	0.073876	1.18581864
Season(June)	0.12012139	0.10799322	1.1123	0.133004	1.12763372
Season(July)	0.02231005	0.09060672	0.2462	0.402752	1.02256078
Season(Aug)	0.17266189	0.10959545	1.5754	0.057577	1.18846421
Season(Sep)	-0.05007565	0.11382841	-0.4399	0.329997	0.95115747
Season(Oct)	0.23554026	0.11176712	2.1074	0.017541	1.26559233
Season(Nov)	0.11816355	0.11237879	1.0515	0.146520	1.12542816
Season(Dec)	-0.04376785	0.10959169	-0.3994	0.344810	0.95717614
Number. of causes	0.03232867	0.01075309	3.0065	0.001322	1.03285692
No Injured	0.00430240	0.00099251	4.3349	7.292e-06	1.00431167
No Killed	0.01822524	0.00319509	5.7041	5.847e-09	1.01839233

Table 3. 11: Deviance, AIC and BIC value for Generalized Negative Binomial Regression

Parameter	Value
Residual deviance	87.0556
AIC	510.6621
BIC	550.7261

Table 3. 12: Model selection

Parameter	PRM	NBRM	GNBRM
Residual deviance	25516	102.4979	87.0556
AIC	25890	551.487	510.6621
BIC	25925.85	589.1943	550.7261

The Generalized Negative Binomial Regression model fit the rate of road traffic crashes data in Kwara state with the minimum AIC (510.6621), BIC (550.7261) and the Deviance value (87.0556).

From the results obtained from generalized negative binomial regression model, month of October, February, May and August are significant in the model while other months are not. The number of causes, number of people killed and the number of injured people have

significant effect on the rate of road traffic crashes in kwara state. The expected rate of road traffic crashes cases is approximately 8 cases per month in kwara state assuming other variables in the model are held constant. The expected rate of road traffic crashes in kwara state is 23.87% higher for the month of February compared to January, assuming other variables are held constant in the model. The expected rate of road traffic crashes in kwara state is 9.54% increase for the month of March compared to January, assuming other variables are held constant in the model. Comparing the month of April to January the expected rate of road traffic crashes is associated with an increase of 1 cases. The month of May has an increase of 18.58% more cases of road traffic crashes compared to January. The expected rate of road traffic crashes in kwara state is expected to be 1 cases higher for the month of June compared to January, assuming other variables are held constant in the model. The month July is associated with an increase in the cases of road traffic crashes rate by a factor of 2.26% compared to January. Comparing January to the month of August is associated with an increase in the rate of road traffic crashes by a factor of 1 cases. Comparing the rate of traffic crashes in the month of to the month of September to January there is a decrease in the cases recorded by 4.88%. The expected rate of road traffic crashes recorded in Kwara state is approximately 1 case higher for the months of Octobers compared to Januarys. Comparing the month of November to January has 12.54% increases in the rate of road traffic crashes cases in Kwara state. The expected rate of road traffic crashes recorded in Kwara state is 4.28% decrease for the month of December compared to January. The month of February and October has high rate of increment in the

rate of road traffic crashes cases. Violation of one or more road traffic rules by motorist and pedestrians increases the rate of road accident by 3.29%. Comparing Poisson, Negative binomial and Generalized negative binomial model, the Generalized Negative Binomial Regression model is the ideal model to fit the rate of road traffic crashes in Kwara state with the minimum AIC(510.6621), BIC(550.7261) and the Deviance value(87.0556).

CHAPTER FOUR

SUMMARY, CONCLUSION AND RECOMMENDATION

4.1 Summary

From the Descriptive statistics results, the highest number of road traffic crashes cases recorded was 44 cases. The summary descriptive statistic as shown in Table 1 revealed that the total observations were 78 while the mean of the number of causes was (6.35) and its corresponding standard deviation was (2.549). Similarly, the number of vehicles involved have the mean value (26.49) with a standard deviation (14.685), number of injured people has a mean value of (65.83) and its corresponding standard deviation was (41.033), and number of people killed has a mean value of (13.08) and its corresponding standard deviation was (10.585). However, the total number of the incident recorded between the various degrees of injury has the mean value of (18.41) with a standard deviation (10.404). One sample Kolmogorov smirnov test was conducted to confirm the distribution of the data. The multicollinearity test was conducted to check if two or more covariate were producing the same information and those producing the same information were removed. An omnibus test was carried out to confirm if the model with no predictor was better fit than a model with predictors. The Poisson regression model was fitted and goodness of fit test was carried out, the main assumption of dispersion was not met since the variance of the dependent variable was greater than the mean and the deviance value (405.020) is

greater than 1 which led to the adoption of the Negative Binomial regression and Generalized Negative Binomial Regression model to remedy the presence of over dispersion. The Generalized Negative Binomial Regression model was confirm to the best fit. From the parameter test carried out under Generalized Negative Binomial Regression, the intercept, month of October, February, May and August are significant in the model while other months are not. The number of causes, number of people killed and the number of injured people have significant effect on the rate of road traffic crashes in kwara state. The expected number of road accident in Kwara state is approximately 8 cases assuming other variables in the model are held constant. The expected rate of road traffic crashes in kwara state is 23.87% higher for the month of February compared to January, while holding the other variables constant in the model. The expected rate of road traffic crashes in kwara state is 9.54% increase for the month of March compared to January assuming other variables constant in the model. Comparing the month of April to January the expected rate of road traffic crashes is associated with an increase of 1 case. The month of May has an increase of 18.58% more cases of road traffic crashes compared to January. The expected rate of road traffic crashes in Kwara state is 1cases higher for the month of June compared to January assuming other variables constant in the model. The month July is associated with an increase in the cases of road traffic crashes rate by a factor of 2.26% compared to January. Comparing January to the month of August is associated with an increase in the rate of road traffic crashes by a factor of 1 case. Comparing the rate of traffic crashes in the month of to the month of September to January there is a decrease in the cases of road

traffic crashes by 4.88%. The expected rate of road traffic crashes in kwara state is 1 case of accident higher for the months of Octobers compared to Januarys, assuming other variables constant in the model. Comparing the month of November to January has 12.54% increases in the rate of road traffic crashes cases in Kwara state. The expected rate of road traffic crashes in kwara state is expected to be 4.28% decrease for the month of December compared to January, assuming other variables constant in the model. The month of April, May, August and October has high rate of increment in the rate of road traffic crashes cases. Violation of one or more road traffic rules by motorist and pedestrians increase the rate of road traffic accident by 3.29%. Most of the road Traffic cases occur most in the first and second quarter of the years Comparing Poisson, Negative binomial and Generalized negative binomial model, the Generalized Negative Binomial Regression model fit the road traffic crashes data in kwara state with the minimum AIC(510.6621), BIC(550.7261) and the Deviance value(87.0556).

4.2 Conclusion

From our results of the analysis carried out, it can be concluded that there is high rate of road traffic accident in the first quarter (Q1) and the second quarter (Q2) of the years. The least cases of road accidents were recorded in the months of Septembers and Decembers. It was also revealed that violation of one or more traffic rules contribute significantly in increasing the road traffic accidents by approximately 3.3%. The Poisson regression, Negative Binomial regression, and Generalized Negative Binomial Regression for the

occurrence of road traffic crashes in Nigeria were considered. On the basis of the AIC values the estimated AIC for PR model (Table 8) is 25890 whereas it is 551.487 for the NB (Table 10) and 510.6621 for the GNBR model (Table 12). The smallest AIC value is that of the Generalized Negative Binomial Regression model. Therefore, the best model for the number of road traffic crashes in Kwara state road safety command is best modeled and described using the Generalized Negative Binomial Regression model.

4.3 Recommendation

Sequel to the results of the findings, I recommend the followings;

1. Driver's license should be made compulsory for drivers to avoid incompetent drivers driving heavy truck on the road.

2. More attention should be paid to the first quarter and second quarter of the years to minimize road accident because accident occur most during this periods.

3. Traffic rules should be made strict on motorist and pedestrians, failure to abide with the rules should result to serious penalty.

4. Elderly people above the age of 50 years should not be allow to drive in the night and in the rain as weather could affect such people which could result to road accident.

5. Traffic inspectors should be appointed at every junction to maintain proper flow of traffic.

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